

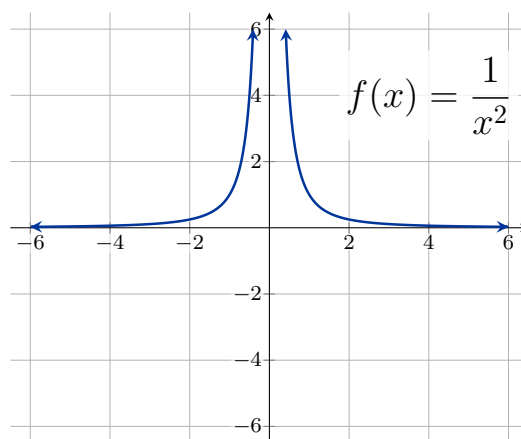
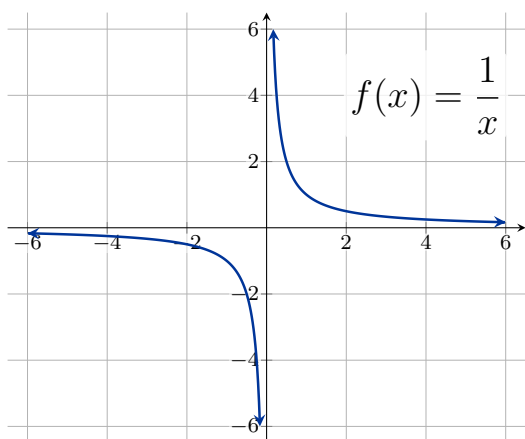
3.7: Rational Functions

Definition. (Rational Function)

A **rational function** is a function of the form

$$r(x) = \frac{f(x)}{g(x)},$$

where $f(x)$ and $g(x)$ are polynomials.



Definition. (Asymptotes)

A **vertical asymptote** of a graph is a vertical line $x = a$ where the graph tends towards positive or negative infinity as the input approaches a from either the left or the right. We write

$$\text{As } x \rightarrow a^-, f(x) \rightarrow \pm\infty \quad \text{or} \quad x \rightarrow a^+, f(x) \rightarrow \pm\infty.$$

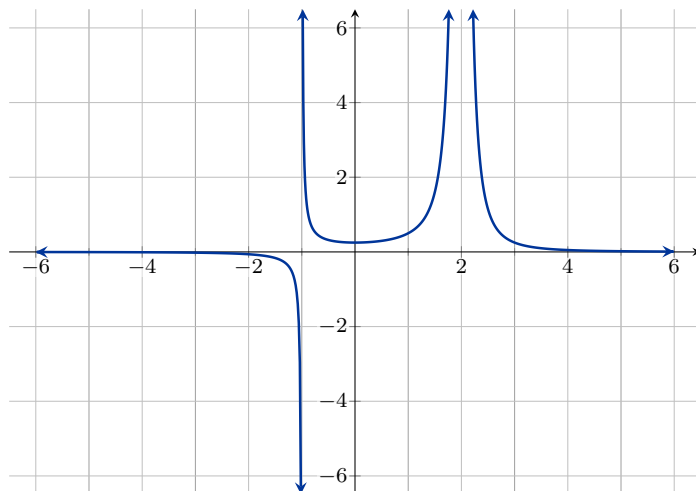
A **horizontal asymptote** of a graph is a horizontal line $y = b$ where the graph approaches the line as the inputs increase or decrease without bound. We write

$$\text{As } x \rightarrow -\infty \text{ or } x \rightarrow \infty, f(x) \rightarrow b.$$

Example. Describe the asymptotes, intercepts, and domain of

$$f(x) = \frac{1}{(x+1)(x-2)^2}$$

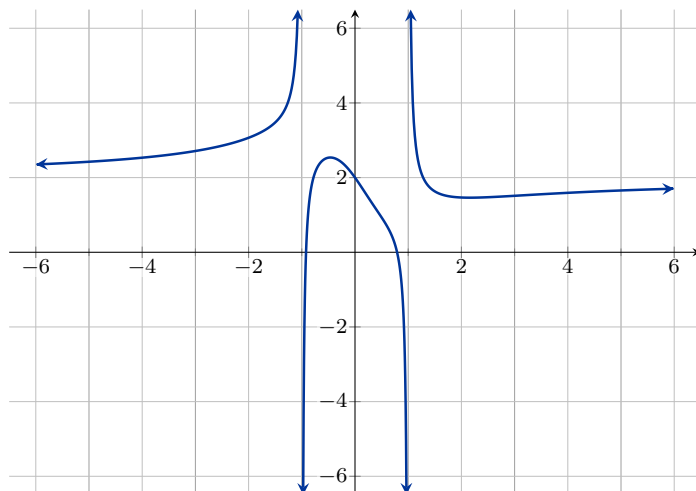
using the graph below:



Example. Describe the asymptotes, intercepts, and domain of

$$f(x) = \frac{2x^4 - 2x^3 + x^2 + 2x - 2}{(x^2 - 1)(x^2 + 1)}$$

using the graph below:



Example. Describe the asymptotes, intercepts, and domain of

$$f(x) = \frac{x^2 - 3x + 4}{2(x - 1)}$$

using the graph below:

